Video Stabilization Algorithm Based on Video Object Segmentation

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Abstract—**The goal of video stabilization mainly is to solve the blurred video caused by the unwanted camera movements. Based on the concept of video object which is proposed by MPEG-4, we apply video object extraction algorithm, simplified SIFT feature point extraction algorithm and motion compensation algorithm to process the video data captured from the fire scene and turn it to a clearer piece of video. Specially, this article proposes an algorithm, which can treat the different video objects respectively based on their value of information and reduce the time wasted on the background region effectively.**

Keywords-fire scene; digital video stabilization; motion compensatio; video object

I. INTRODUCTION

Due to lower costs of the computer equipments and the progress in video image processing technique, more and more real-time video applications are applied to the fire-fighting field in recent years. The blurred video caused by the camera unwanted movements, leads to a new problem which is how to get the stabilized video.

In order to solve this problem, the video data analysis and feature extraction is necessary. In traditional approaches, the video stabilization technique processes video dominant objects containing important information and unimportant background region in the same way. This approach has been widely applied to various environments, but it cannot meet high real-time requirements in the fire-fighting field.

This paper presents a new video stabilization algorithm based on video object segmentation. We design and implement a video stabilization system according to the fire scene video features, which refers to traditional video stabilization technique. This system just produces a smaller delay to complete the video stabilization task. The algorithm includes three parts: video object extraction, feature point extraction and matching, and video image compensation. We have improved the sophisticated algorithms to adapt to the features of the fire scene better, so as to provide more valuable information for the fire scene command post.

The paper is organized as follows. Section Ⅱ mainly analyze features of the video captured from the fire scene. In Section Ⅲ we propose a new video stabilization algorithm. Our experiments validate the performance of the algorithm in Section 4. Finally, we conclude the paper in Section Ⅴ.

II. FEATURES OF THE VIDEO CAPTURED FROM THE FIRE SCENE

In the video capture from the fire scene, video data mainly come from a portable wireless camera carried by the firefighters. So, when the locations of firefighters are different, the contents of the video data recording will be different too. When the firefighters are outside the fire scene and hold fire hose spraying of water to put out the fire, they basically do not move and face the fixed direction. In this case, both the system operation time and the delay are short.

In the second case, when firemen do the modest movement, such as walking under the normal speed, slowly rotating and so on, and scene captured by camera will change with them. This change is quite mild, so the computing power of the video stabilization system is required very low.

In the last case, during the process of firefighters rushing into the fire scene and performing in-house fire extinguish or rescue operation, camera will fiercely rock and video image will appear bumpy, fuzzy and unstable. At this time we need to use video stabilization system to process a large number of video images in order to remove the image jitter and save the information contained in the video. It can provide a reliable basis for the fire scene command post to make correct decision. Based on the above analysis of basic features of the original video, we can obtain the following two conclusions: (1) The random movement of the portable wireless camera will cause the video image to varying degrees of jitter and vague. Based on this assumption, the maximum extent to ensure enough information to be contained in the video image, the video stabilization system needs to achieve a certain balance between the system latency and the video image quality to ensure the overall real-time property and effectiveness. (2) The quality of camera itself and the complex fire scene may result in a large number of video noises. In this case video stabilization system should reduce video noise and provide a uniform and stable video data for subsequent processing steps.

III. A NEW ALGORITHM

This paper proposed a video image stabilization algorithm which includes three parts: video object extraction, the video image feature point extraction and matching, and the video motion estimation and compensation. The structure of the system as shown in Fig. 1.

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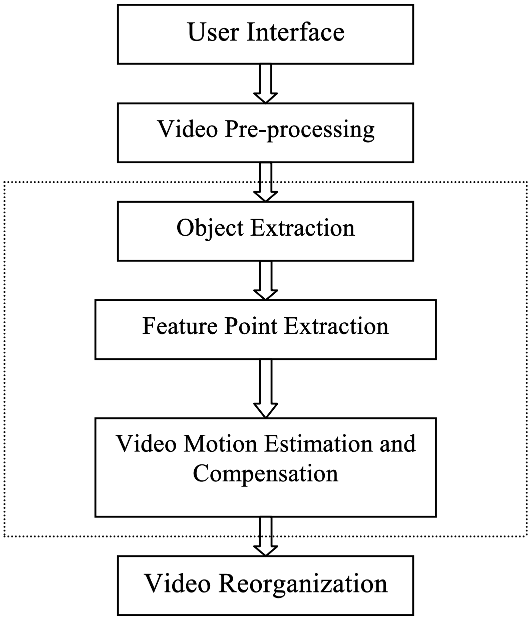

Figure 1. The structure of the system

A. Video Object Extraction

MPEG-4 video coding standard provides object-based functionalities by introducing the concept of Video Object Plane (VOP). With the extraction of video objects and allocation of different number of bits or different frame-rates for different objects, the standard can support object-based scalability, which is very useful in our system. Video object extraction algorithm is used to distinguish between video dominant objects and background region. In order to simplify users' operation, our algorithm pre-stores several important edge information of the object in the fire scene, such as body, tanks, gas cans and so on. This paper uses the inter-frame affine, which gets information from the previous frame to make up for the current one and determine the correlation between them, to identify the video dominant objects in video image sequences. We assume gray-scale mask of previous frame as T_{i-1}, bi-level image of current frame as B_i, affine transformation as O_{prj} and inter-frame displacement as D_i. Through the following steps, we can obtain the current frame of the gray-scale mask T_i[1,2,3].

1) Inter-frame affine: Any coordinates in the current frame $P(x,y)$, can be determined by the location in the previous frame $P'(x',y')$, and $P'=P-Di$. If P' is in the previous frame and gray-scale difference is less than a certain threshold, then that point P belongs to a new object.

2) Obtain the current frame of the gray-scale pattern: Obviously, T_i is equal to B_i and O_{prj}.

3) Access to video object: Through the T_i and T_{i-1}, we can obtain two corresponding relationships between the pixels, determine the contours of the video dominant objects, and distinguish the video dominant objects from the background. After edge matching, video dominant objects and background are processed separately.

This paper uses the overlap region, the previous and current frames, and the edge of video objects of the current frame[1,3]. This approach can solve shadow problems between rigid video objects such as oil drums stacked together and problems between non-rigid video objects such as humans side by side. On the other hand, we needn't do anything for the background.

Finally, the video object and background are stored separately, while their location information in the image is also recorded.

B. Video image feature point extraction and matching

The main role of feature point extraction algorithm is to extract the stable feature from video image. SIFT(Scale Invariant Feature Transform) consists of four major stages: (1) Scale-space peak selection; (2) Key-point localization; (3) Orientation assignment; (4) Key-point descriptor. We construct the Gaussian Pyramid and search for local peaks (termed key-points) in a DoG (Difference of Gaussian) Pyramid. In the second stage, candidate key-points are localized to sub-pixel accuracy and eliminated if found to be unstable. The third identifies the dominant orientations for each key-point based on its local image patch. The assigned orientation, scale and location for each key-point enables SIFT to construct a canonical view for the key-point that is invariant to similarity transforms. The final stage builds a local image descriptor for each key-point, based upon the image gradients in its local neighborhood.

Video image samples in two-dimensional space are fixed, so we sample peaks in a variety of scale space[4,5,6]. The system uses a fourth-level Gaussian pyramid, each of which contains four layers. Each layer is generated by the Gaussian convolution of a previous layer, and the generated formula is shown in the Equation (1).

$$L(x,y\sigma)=G(x, y\sigma)*I(x,y). \tag{1}$$

In order to reduce the complexity of the algorithm and improve the system's real-time property, we use simplified SIFT pyramid structure, which deletes the top layer in each level. In the DoG Pyramid, the top-layer image of each level has been deleted[7]. We use the corresponding peaks comparison, which is between the first and second layer image in upper level, to replace the comparison, which is between the top two layer images in current level. The bottom layer image of each level still does not extract the key-point. So we decreased a Gaussian convolution in each level.

The final stage (key-point descriptor) of SIFT algorithm builds a representation for each key-point's location, rotated pixels in its local neighborhood. The standard key-point descriptor used by SIFT is created by sampling magnitudes and orientations of the image gradient in the patch around the key-point, and building smoothed orientation histograms to capture the important aspects of the patch. A 4*4 array of histograms, each with 8 orientation bins, captures the rough spatial structure of the patch. Then this 128-elemnet (4*4*8) vector is normalized to unit length and threshold to remove elements with small values.

The standard SIFT feature vector is complicated for our system. We don't sample the magnitudes and orientations of the image gradient in the patch around the key-point, but according to the pixel magnitudes and orientations of the image gradient in the patch around the key-point, we calculate feature vectors directly.

In our system, we have image sequences of the object in the fire scene. Two key-points corresponded to the most similar feature vectors in each image, form a matched pair. By calculating the Euclidean distance ratio of feature vectors, we determine whether the corresponding key-points match.

C. Video motion estimation and compensation

Motion estimation is the process of determining motion vectors which describes the transformation from one 2D image to another; usually from the adjacent frames in a video sequence. Our system uses LSS (Line-Square Search) motion estimation algorithm based on the motion vectors and SAD (Sum of Absolute Difference) characteristics in the spatial distribution. This algorithm can be used for different regions in different ways. LSS algorithm performs the line search for the unimportant area to reduce the computation complexity. For the important search area in which optimal points are existed, a square search pattern consisted of 9 checking points is used to carry out the refined search. So there are two kinds of search pattern, Square search pattern and linear search pattern. This idea is very suited to our system. Square search pattern has nine checking points called inner pattern and eight analog points called outer pattern. Refined search use inner pattern and select the linear search direction, which adopt outer pattern. Fig. 2 [8] indicate inner and outer pattern.

Through motion estimation algorithm, the video stabilization system has differences between the adjacent video image sequences, and obtains the corresponding motion vectors. Based on the information, the system can process motion compensation and video image reconstruction.

Motion compensation steps are as follows: 1) Video image will be split into static parts and moving parts. 2) Obtain the previous frame image data according to motion vectors. 3) Through the use of prediction filter, obtains the forecast difference block of the previous frame image. 4) Combine static parts and the forecast difference block to regenerate a new image.

IV. EXPERIMENTAL DATA

To verify the effectiveness of the system, this article selected 6 different videos shot in different environments. The performance experiments are in half-pixel search and 8×8 sub-block case. Experimental results are shown in Table Ⅰ.

Indicated by the data in Table I, the video stabilization system processing delays can meet the requirements of the fire-fighting field. However, in some complex environments, because of scene light, shadow and other factors, there are some problems about video object extraction and the image distortion. These issues need to be researched in the future.

V. CONCLUDES

We design and implement the video stabilization system by referring to lots of video image processing literatures. Compared with existing algorithms, the performance of the algorithm proposed by this paper has been greatly enhanced and improved in some areas.

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TABLE I. EXPERIMENTAL RESULTS OF THE VIDEO STABILIZATION

ID	Length(m:s)	Light conditions	Collection Status	V/E*	Average processing time (ms)	Whether distort
1	7:22	Good	Slow Step	5/5	270.3	NO
2	6:43	Strong Light	Slow Step	5/5	254.7	NO
3	11:49	Good	Quick Step	7/6	276.2	YES
4	5:16	Dark	Slow Step	3/3	260.4	NO
5	7:19	Torch Lighting	Quick Step	4/4	259.5	NO
6	10:07	Good	Run	6/6	266.3	NO

* The number of video objects/ the number of video object extractions is set as V/E.

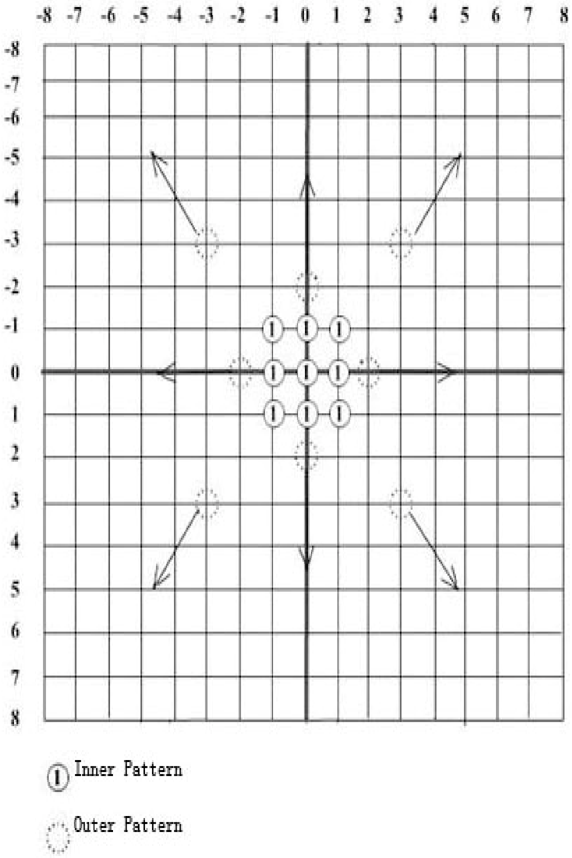

Figure 2. The inner pattern and the outer pattern